

Philip Baah Afriyie

AI/ML Engineer · MLOps

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Summary

Machine Learning Engineer with three years of production experience building and deploying agentic AI systems end to end — from fine-tuning speech and vision models to shipping the event-driven infrastructure that serves them. I work across the whole stack: LangChain and LangGraph agents, retrieval-augmented generation, real-time voice pipelines, OCR document extraction, and the Kafka, Redis, Docker and AWS plumbing that keeps them running. Comfortable owning a system from a research notebook through to a load-balanced production deployment.

Experience

Machine Learning Engineer — Ayadata GH

Nov 2023 – Present · Accra, Ghana

- **Real-time voice AI survey platform.** Designed and shipped an unattended IVR system that conducts spoken telephone surveys in a local Ghanaian language: speech recognition, machine translation, answer classification, dynamic question routing, and speech synthesis in one live pipeline. Built two front ends over a shared survey engine — a LiveKit voice agent for real-time calls with energy-based speech segmentation, and a FastAPI service for batch processing. Deployed behind Gunicorn and nginx with disk-backed session state so any worker can resume any call in progress.
- **Agentic performance analytics platform.** Built a production dashboard that aggregates business and operational metrics into real-time risk scoring, flags employees at risk of a Performance Improvement Plan, and generates LLM-written performance insights for HR and management. Added a multi-channel notification service (Slack, email, push) with severity-based routing, and role-based access control separating admin, HR and manager visibility.
- **Agentic document extraction.** Developed an end-to-end pipeline that ingests raw PDFs and scanned images and converts them into structured PostgreSQL records via OCR, with downstream agents running analytics on the extracted data to generate reports automatically and cut manual administrative work.
- **Event-driven agent infrastructure.** Deployed agentic RAG systems on LangChain and LangGraph for autonomous context retrieval, orchestrated with Kafka for high-throughput event processing and Redis for conversational state and caching.
- **Speech model fine-tuning & serving.** Fine-tuned Whisper (ASR) and a bidirectional NLLB model on real, code-switched farmer speech — English brand names and numerals mixed into Twi — then deployed both behind a low-latency NVIDIA Triton serving layer with a custom latency/throughput benchmarking dashboard.
- **Computer vision for agriculture.** Engineered object detection models for automated crop counting, reaching 91.8% precision on banana trees with MMDetection and 94% precision on palm trees with Ultralytics YOLOv8, to improve harvest planning.

Selected Projects

Repositories are private; I'm glad to grant access on request.

Voice AI Survey Platform — real-time conversational IVR

Python, FastAPI, LiveKit, Claude on AWS Bedrock, Gunicorn, nginx
github.com/Philips-DR/voice-ai-survey-platform

- A phone-call-style survey agent that understands free-form spoken answers and branches to the next question with no human operator on the line, on a declarative question graph with conditional branching and a time-based rule that shortens an overrunning call.
- Five answer-parsing strategies selected per question type — negation-aware keyword matching, fuzzy multi-option scoring, regex quantity extraction, live LLM entity extraction, and post-call theme coding — so open-ended answers never add latency; every turn logged for replay and audit.

docu-ai — markdown to Google Docs compiler

TypeScript, Google Docs API, OAuth 2.0, Vitest github.com/Philips-DR/docu-ai

- Compiles a folder of Markdown into a single properly formatted Google Doc — real Docs styling, syntax-highlighted code blocks, tables, embedded images, one tab per chapter and a clickable table of contents — with zero markdown residue in the output.
- Layered pure-function architecture (parse, plan, emit, verify) with golden request snapshots, so the whole compiler is testable without network access or credentials.
- Verification-first: a residue lint reads the built document back and asserts no unconverted markdown survived, and font application is proven by PDF export rather than trusted from the API response.

Akan-English Speech Pipeline — Fine-Tuning & Production Serving

Hugging Face Transformers, PyTorch, NVIDIA Triton Inference Server, FastAPI, MLflow, Terraform

- **STT:** Fine-tuned Whisper on real, code-switched farmer speech (English product names and numbers spoken mid-sentence inside Akan), reaching roughly 7% word error rate — after building a text-normalization pipeline to clean inconsistent Akan orthography across thousands of transcripts.
- **MT:** Fine-tuned a bidirectional Akan-English NLLB-200 model to a BLEU score of 81, tracked in MLflow alongside chrF++ and ROUGE.
- **TTS & serving:** Trained a custom Akan text-to-speech voice, then deployed all three models behind an NVIDIA Triton server fronted by a FastAPI gateway (auth, rate limiting) and provisioned with Terraform/Ansible — validated with a dashboard benchmarking latency and throughput.

Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **AI and LLM systems:** LangChain, LangGraph, agentic RAG, retrieval-augmented generation, prompt engineering, Claude on AWS Bedrock, Hugging Face Transformers, PyTorch, model fine-tuning, OCR (Tesseract, PaddleOCR)
- **Machine learning:** Scikit-learn, MMDetection, Ultralytics YOLOv8, object detection, NLP, speech AI
- **MLOps and infrastructure:** Docker, Kubernetes, NVIDIA Triton, Terraform, Ansible, Kafka, Redis, MLflow, DVC, Gunicorn, nginx
- **Cloud and data:** AWS (EC2, S3, Lambda, Bedrock), PostgreSQL, FastAPI, REST APIs, OAuth 2.0
- **Tools:** Git and GitHub, Power BI, QGIS, VS Code

Education

BSc. Information Technology — BlueCrest University College

Accra, Ghana · CGPA 3.63

Certifications

Natural Language Processing Specialization (DeepLearning.AI, Coursera: Classification and Vector Spaces, Probabilistic Models, Sequence Models, Attention Models) · **AWS Cloud Technical Essentials** (Coursera) · **Conversational AI Expert Author** (DRUID AI) · **Foundations: Data, Data, Everywhere** (Google Data Analytics, Coursera)